An Empirical Study on Tea Leaf Disease Detection on Utilizing Diverse Leaf Image Features

Pankaj Pratap Singh, Dristi Nayana Borah, Jutika Basumatary, Jyotibhushan Hazarika, Pamee Brahma, and Shitala Prasad

Abstract In this study, a machine learning (ML) based method is proposed for the early detection of plant leaf diseases. Tea leaf disease detection plays a crucial role in enhancing agricultural productivity and ensuring the health of tea plantations. In this study, a comprehensive methodology for tea leaf disease detection utilizing image processing (IP) and machine learning (ML) techniques is presented. Our approach involves the extraction of diverse features from tea leaf images, encompassing color, texture, shape, and size attributes to capture a wide range of characteristics indicative of disease presence. These features are then utilized to train and evaluate a classification model capable of accurately distinguishing between healthy and diseased leaves. Experimental findings underscore the efficacy of our proposed approach in accurately identifying tea leaf diseases, thus facilitating the advancement of plant disease management strategies within the tea cultivation domain, enhancing agricultural productivity, and ensuring the health of tea plantations.

Keywords Tea leaf features · Tea leaf disease classification · Segmentation · Image processing · Machine learning

1 Introduction

Plant diseases pose significant challenges to agriculture and impacting crop yields and economic stability. These diseases are caused by various factors, such as environmental factors and pathogens, including fungi and bacteria. Symptoms of these plant diseases range from wilting and discoloration to deformities and decay, making their identification and management crucial for agricultural sustainability. Tea, as one of the world's most consumed beverages, plays a vital role in economics and cultures,

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providing economic stability and cultural significance to communities worldwide. But the various tea diseases always pose a threat to tea cultivation. Climate change, environmental factors can continuously damage tea leaves, resulting in significant losses and economic hardship for farmers. Therefore, the importance of early detection and diagnosis of tea leaf diseases becomes really important. Timely detections are needed to reduce the impact of diseases and protect tea crops from these damages. Manual detection methods are often time-consuming and require intense labor. The urgent need for effective and reliable detection solutions has led to the development of automated systems with advances in computer vision (CV) [1, 2].

This study aims to develop an automated tea leaf disease diagnosis approach that can accurately detect and diagnose diseases in tea leaves. Through testing and validation, our goal is to demonstrate the efficiency and effectiveness of our solutions in overcoming the challenges faced by tea farmers and implement them successfully. Tea diseases exhibit a wide range of symptoms, including discoloration and deformity. These symptoms can vary depending on various factors such as disease type, environmental conditions, and stages of development. Identifying these subtle traits requires systems capable of detecting nuanced differences in color, texture, shape, and structure. The primary objective is to develop a Plant Disease Detection System capable of accurately classifying plant leaves based on their health status using machine learning (ML) algorithms and image processing (IP) techniques. By extracting meaningful features relevant to disease identification, the objective is to enhance the accuracy and efficiency of detection methods.

2 Literature Survey

The goal of the research is to use machine learning and deep learning (DL) techniques to evaluate and examine different approaches for plant disease identification. These different concepts, like Random Forest, Convolutional Neural Network (CNN), and IP and CV, have been studied [3]. Random Forest is an ensemble learning method that constructs multiple decision trees during training and outputs the mode of the classes (classification) or the mean prediction (regression) of the individual trees. It can be applied to tea leaf disease detection by utilizing features extracted from leaf images. CNNs are machine learning models that are specially designed for structured data like images. CNN can be applied to tea leaf disease detection by directly processing images [4, 5]. IP and CV can be applied to analyze digital images. These methods can detect and classify based on features extracted from images.

Some researchers work on plant disease detection (see Table 1). Every research study examines a different facet of automated illness diagnosis, from the use of cutting-edge methods like deep learning to the accuracy and efficiency of algorithms. Singh and Misra [6] worked on an automated plant leaf disease detection system based on a genetic algorithm and machine learning. The system utilized a dataset of leaf images and achieved promising results with an SVM classifier [7].

Table 1 State of the art related to plant disease detection

Author and year	Objective	Dataset used	Approach	Result	Limitations
Singh and Misra [6] and (2016)	Automate plant disease detection, validate efficiency	Plant leaves	Genetic algorithms (GA), ML	95.71% accuracy with SVM classification	Lack of real-world testing and challenges in classifying complex diseases due to simplistic techniques and limited dataset diversity
Mohanty et al. [8] and (2016)	Enable deep learning for rapid plant disease detection	Plant Village	IP and ML	99.35% accuracy in plant disease diagnosis, with a range of 85.53–99.34%	Require diverse training data for realistic settings
Joshi et al. [9] and (2022)	Develop efficient plant disease detection using CV and ML	Plant Village	IP and ML	Achieved 95% accuracy on potato leaf images	Lacks scalability analysis and real-world applicability beyond its dataset
Demilie [10] and (2024)	Compare DL and ML for plant disease detection	Plant Village	Focuses on DL and ML	CNN accuracy and early detection's significance	Insufficiently addresses drawbacks and scalability concerns

Mohanty et al. [8] carried out a project on deep learning for rapid plant disease detection using image processing and received 99.35% in plant disease diagnosis. Joshi et al. [9] developed an efficient plant disease detection system using computer vision and machine learning, including digital image processing. The proposed system demonstrated high accuracy in detecting diseases such as blister blight and gray blight. Demilie [10] explored and compared deep learning and machine learning for plant disease detection using CNN and achieved good accuracy and early detection's agricultural significance.

3 A Proposed Model

In response to the crucial need for early and correct detection of tea leaf diseases to guard crop yield and quality. The dataset of tea leaf images includes seven diseased types (Anthracnose, algal leaf, bird eye spot, brown blight, gray light, red leaf

spot, white spot) and one healthy type. The images are preprocessed to remove the background, resize, enhance brightness, and apply normalization. Four features are extracted, including color (calculating each channel of Hue, Saturation, and Value), shape and size (measuring the size of each shape and extracting image dimensions, including contours, area, width, and height), and texture (using Local Binary Patterns or LBP).

3.1 System Architecture

The tea leaf disease detection architecture classifies the disease images based on their color, texture, and size (see Fig. 1). The steps are as follows:

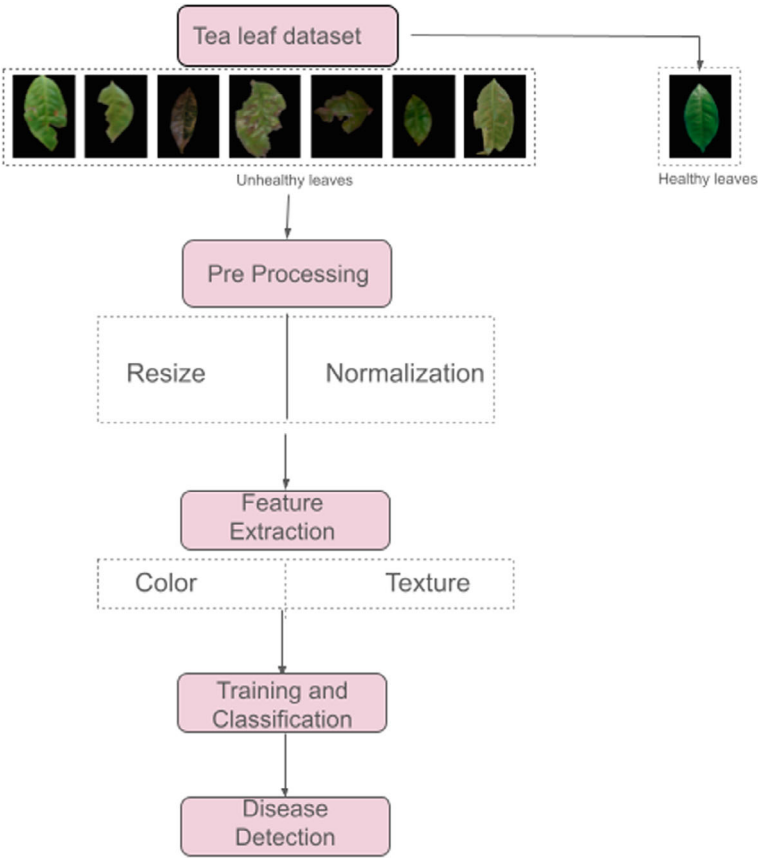


Fig. 1 System architecture

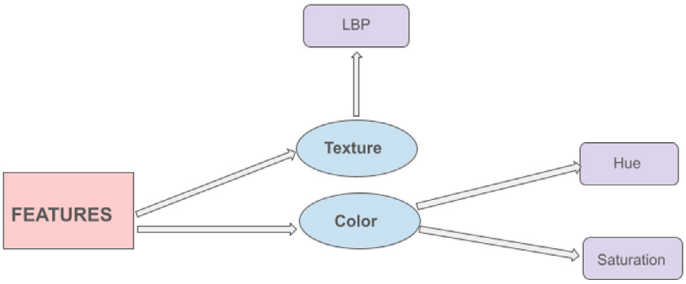


Fig. 2 Feature extraction from the tea leaf images

- Preprocessing: The dataset is preprocessed to remove any noise and background, enhance the brightness to improve the quality, and then apply normalization.
- Feature Extraction: In this step, the system identifies and extracts relevant features from the preprocessed images, which are used to distinguish healthy and unhealthy tea leaves.

The methods for extracting features are as follows:

- For texture, using methods like Local Binary Patterns (LBP).
- For color, calculation of each channel (Hue, Saturation, and Value).
- Training: The extracted features are then used for training. The dataset is split into three parts: Training, testing, and validation.
- Model Selection and evaluation: Selection of models based on evaluation metrics. The model learns to classify the tea leaves as healthy or unhealthy based on the extracted features.
- Disease Detection: The system can now detect diseases in new tea leaf images. It takes an image as input, pre-processes it, extracts features, normalizes them, and then uses the trained model to classify whether the leaf is healthy or unhealthy.

The system takes a tea leaf dataset as input, which undergoes pre-processing tasks like resizing, color, and texture analysis. Then, features are extracted, normalized, and classified into healthy or unhealthy leaves for training and disease detection purposes. The extracted features using image processing from the tea leaf images are shown in Fig. 2. It mentions several features that can be extracted, including Local Binary Patterns (LBP), texture, hue, color, and saturation.

4 Implementation and Results

In order to deliver precise and trustworthy classification findings, a strong framework for tea leaf disease identification is established. The procedure started with the collection and preparation of a heterogeneous dataset that included tea leaves images in good condition, with those that are diseased shown in Fig. 3.

Fig. 3 Tea leaf dataset

4.1 Dataset Details

The tea leaf disease dataset is taken from Kaggle, which has 7 types of diseased and one type of healthy leaves categories [11]. Total images with the type of disease are shown in Table 2.

Table 2 Healthy and diseased leaf dataset

Name of the leaf	No. of image
Anthracnose	100
Algal leaf	113
Bird eye spot	100
Brown blight	113
Gray light	100
Red leaf spot	143
Healthy	74
White spot	142

4.2 Feature Extraction

After that, a multipronged strategy is used for feature extraction that included form descriptors, texture analysis using methods such as Local Binary Patterns (LBP) in Fig. 4, and color-based features visualizing Hue (H) and Saturation (S) channels are describing color properties of an image as shown in Fig. 5. The shape and size based features extraction with bar plots depicting maximum area, height, width in Fig. 6.

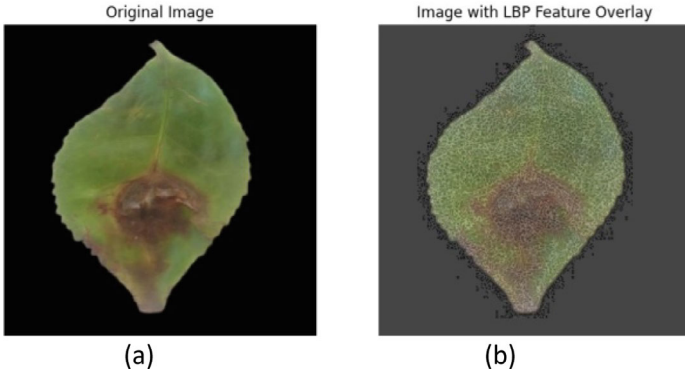


Fig. 4 **a** Input plant image; **b** output LBP feature overlay image

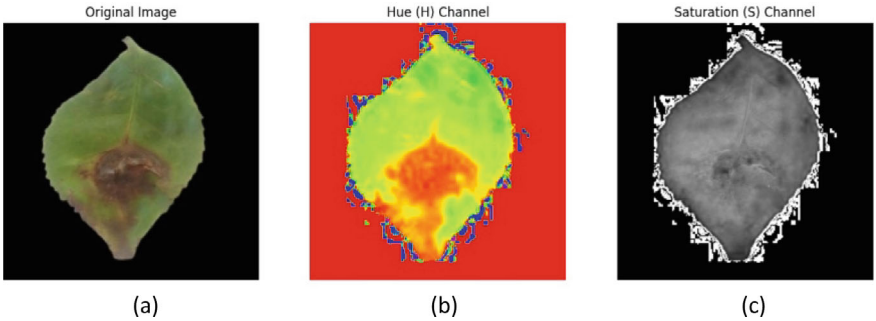


Fig. 5 **a** original image; **b** Hue (H) channel of input image; **c** Saturation (S) channel of input image

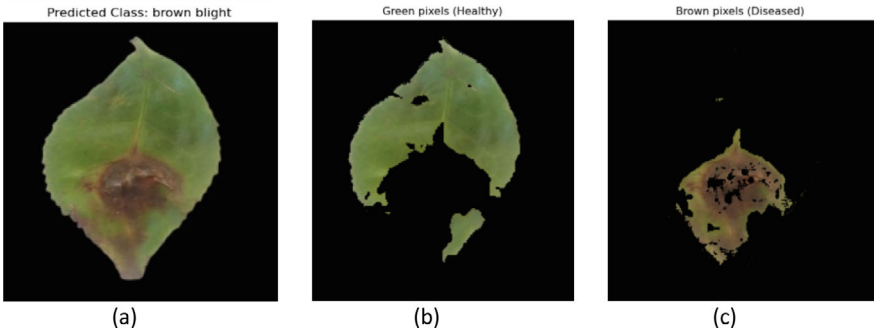


Fig. 6 Predicted brown blight disease along with green and brown pixels

4.3 Results and Analysis

Using an extensive feature set that captured the key elements of tea leaf images. Some machine learning models are trained and assessed. The model's optimum performance is verified by cross-validating our evaluation and fine-tuning the hyper-parameters. This analysis of the input image using various ML models and IP techniques has produced informative results. Firstly, the model accurately predicted the class of the plant pathology, distinguishing between healthy and diseased states.

In this instance, the plant was identified as exhibiting signs of brown blight, Anthracnose, Healthy, and Red Leaf Spot, demonstrating the model's capability to recognize specific pathologies based on visual cues (see Figs. 6, 7, 8, and 9). Additionally, color thresholding in the HSV color space is utilized, where the image is segmented to isolate healthy green regions from diseased brown areas. This segmentation enables precise identification and localization of affected areas, aiding in targeted management strategies for plant health output, where an LBP feature has been applied to highlight certain aspects of the plant. Its overlay creates a texture-based representation of the plant, enhancing its details. It can be clearly seen that some of the diseases are not correctly predicted in Fig. 10.

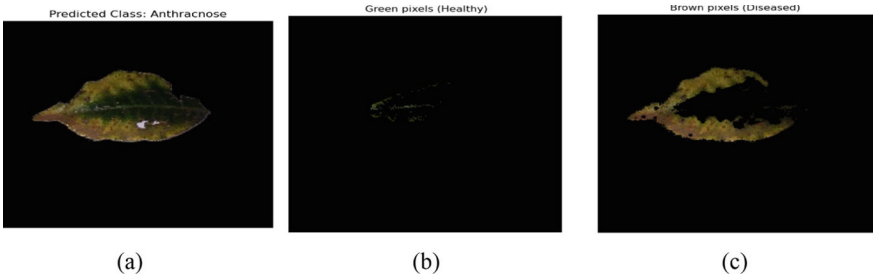


Fig. 7 Predicted anthracnose disease along with green and brown pixels

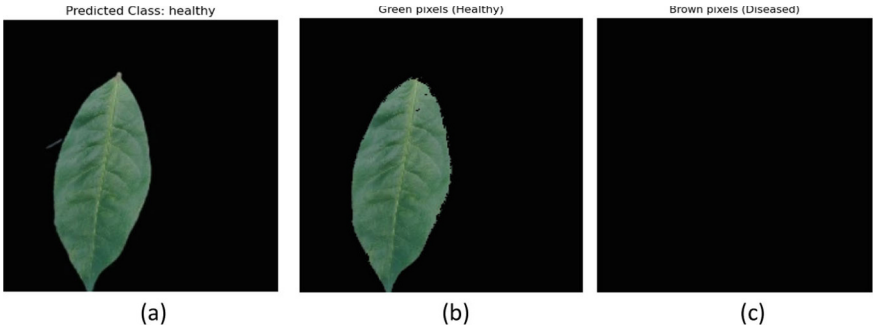


Fig. 8 Predicted healthy leaf along with green and brown pixels

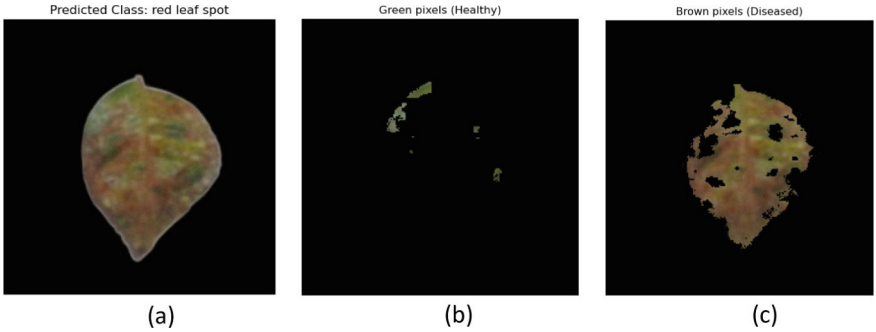


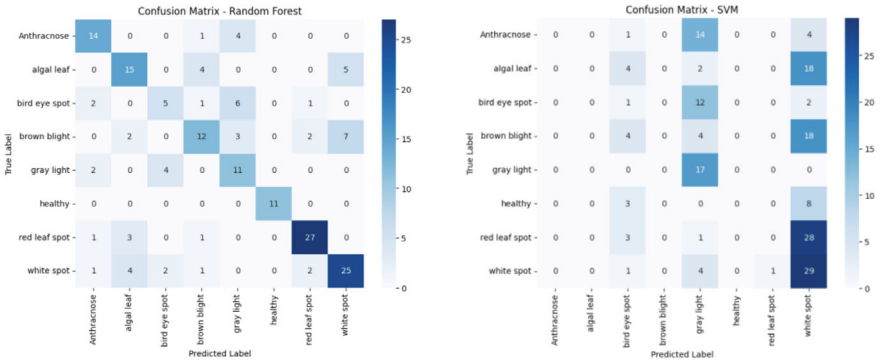
Fig. 9 Predicted red leaf disease along with green and brown pixels

Fig. 10 Incorrect prediction of some diseases

The proposed model's performance is evaluated on a test dataset using a random forest classifier. Data are collected on both true and predicted labels. Quantitative analysis is shown in Table 3. A heatmap is a visualization of the confusion matrix. This heatmap offers a clear representation of the model's classification performance, showing the frequency of correct and incorrect predictions across disease classes (see Fig. 11a, b). Each cell indicates the number of instances where a true label was predicted as a specific class, aiding in identifying areas for model refinement. A comparative analysis among different classifiers is shown in Fig. 12.

Table 3 Quantitative analysis using the different metrics

	Precision	Recall	F_1-score	Support
Anthracnose	0.70	0.74	0.72	19
Algal leaf	0.62	0.62	0.62	24
Bird eye spot	0.45	0.33	0.38	15
Brown blight	0.60	0.46	0.52	26
Gray light	0.46	0.65	0.54	17
Healthy	1.00	1.00	1.00	11
Red leaf spot	0.84	0.84	0.84	32
White spot	0.68	0.71	0.69	35



(a) (b)

Fig. 11 **a** Heatmap visualization of random forest, **b** Heatmap visualization of SVM

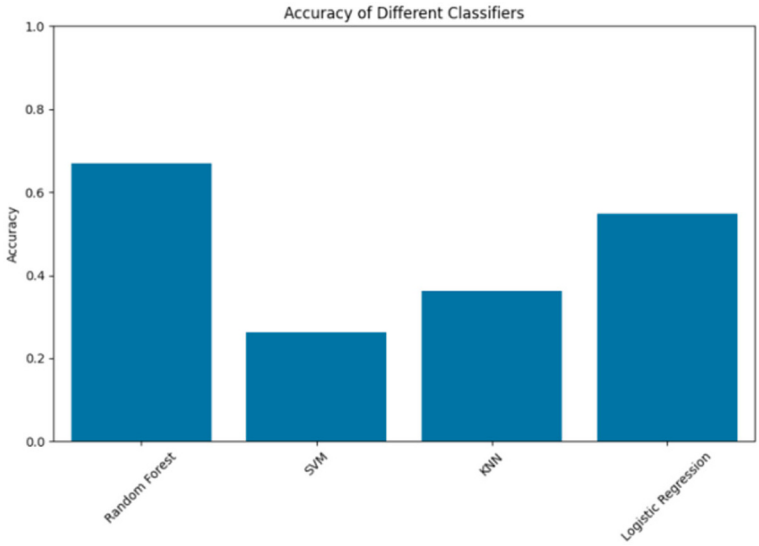


Fig. 12 Quantitative analysis (accuracy) of different classifiers

5 Conclusion

Our research on the identification of tea leaf disease using extracted features offers a potential remedy for the difficulties tea growers encounter in keeping an eye on and controlling plant health. The proposed system can recognize and diagnose illnesses that impact tea plants by combining cutting-edge image processing methods and machine learning algorithms. Our method provides a comprehensive understanding of tea leaf health by extracting and analyzing various aspects, including color, shape, and texture. This allows for prompt intervention and efficient disease control techniques. Our approach can help as a successful application show how it may effectively support improving tea farming techniques, resulting in higher yields, lower losses, and sustainable tea production. These findings showcase the effectiveness of machine learning and image processing in agricultural applications, particularly in the early detection and management of plant diseases. Such approaches hold promise for enhancing crop yield and sustainability by enabling timely interventions to mitigate the impact of diseases on agricultural production.

The future scope of work will encompass other important areas for more research and development. First and foremost, a larger range of tea plant varieties, disease kinds, and environmental factors will be included in the dataset. This will improve the system's efficacy in a variety of agricultural environments by strengthening its resilience and generalizability. In order to improve the precision and effectiveness of illness diagnosis, the integration of other characteristics and sensors, such as spectral imaging and Internet of Things devices, will also be investigated. Since it will evaluate and implement the approach in actual tea farming operations and

guarantee its applicability and practicality in the field, cooperation with tea farmers and industry stakeholders would be crucial.

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